

PROJECT REPORT · DESENSITISED EDITION

Consolidating Legacy Contract Records with an LLM Assistant

Rights Verification and Governance Design
for a Family-Held Property Portfolio

Problem & Constraints · Corpus Analysis · Solution Design · AI-Assisted Workflow
Evidence Grading · Risk Register · Outcomes & Limits

AUTHOR

Zou Zhihua

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Confidentiality and desensitisation

The cover icon is a generic pictogram of a building type. It is not the project's logo, photograph or drawing. Because parts of the underlying project are confidential, this edition has been desensitised: names, identity data, addresses and place names, monetary amounts, ownership ratios, contract wording, parcel identifiers, scans and stamps have been removed or generalised.

What remains is real but presented at a level that cannot identify the parties: the problem structure, the workflow, the deliverable design, and countable facts such as page, clause and checklist-item counts.

1 Background and Objectives

1.1 Problem Statement

Between 2008 and 2018 a family accumulated several property interests in a coastal district of southern China through cooperative-construction agreements on collectively owned land. In each agreement a local landholder supplied the plot and an investor funded construction. Because such buildings generally cannot be registered to the investor, the investor's position rests on contract and on the landholder's cooperation rather than on a title certificate.

By 2026 this position was difficult to act on for four reasons. First, the evidence was a paper trail of about ten agreements, applications and acceptance records, plus supporting identity and land-registration papers, in total 39 scanned pages across four parcels; several pages were handwritten and some were incomplete. Second, the succession status of one contributing party was unverified. Third, there was no single entity to collect income, issue invoices or hold records. Fourth, a proposed split of interests circulated in working drafts without any documentary source. The task was to turn this into a position that a family and its advisers could reason about, and to say precisely what was and was not yet proven.

1.2 How the Arrangements Work

Every agreement in the corpus follows one commercial pattern. Table 1 groups the clause families that recur, because the solution had to be consistent with them.

Table 1: Clause families recurring across the source agreements

Family	What the agreements provide	Consequence for the solution
Land supply	Landholder supplies a village-allocated plot and warrants it is lawfully held, unencumbered and sealed by the village committee	Validity rests on collective approval; the plan must not bypass the collective
Financing	Investor funds construction in full, in cash	Contribution must be evidenced by payment records, not by the wording of the agreement
Allocation	Landholder takes lower floors or a fixed area share; investor takes upper floors; stairs are shared	Entitlements are area-based, so later compensation follows area
Registration	Where separate registration is impossible, the building is filed in the landholder's name ("loan-name" registration); landholder must cooperate if policy later allows division or transfer	The investor's position is contractual; a management entity cannot cure title
Expropriation	Compensation follows the area held; the landholder must pass the investor's part on within days	Future proceeds need an agreed routing
Remedies	Defaulting landholder pays a multiple of assessed value; investor may reclaim the building; investor exit forfeits deposit	Defines leverage and downside for each side
Succession	Legal heirs of either side remain bound and may not dispute the agreement	Justifies verifying heirs before anything is signed
Formalities	Several originals, identity copies, fingerprints, witness and guarantor lines	Explains why identity papers fill 7 of the 39 pages

1.3 Objectives

The work had four objectives:

1. reconstruct one source-cited picture of who holds which rights, from about ten heterogeneous documents;
2. separate what the documents *prove* from what people *assume*;
3. design an executable, ordered plan (heir verification, written confirmation of contributions, a management entity) suitable for review by licensed counsel;
4. keep every AI output labelled as a working draft, never as legal advice.

1.4 Constraints, Role and Boundaries

The main constraints were the absence of title certificates, land-use rules that limit transfer and registration, sensitive personal data in every source file, and one analyst coordinating with family members. The author acted as analyst and coordinator: collecting the originals, directing the LLM assistant, verifying every output against the scans, deciding grades and sequencing. Legal advice, notarisation, tax filing and any registration step were designated for licensed professionals and local authorities and were not performed by the author.

1.5 Disclosure Scope

This report is a *desensitised edition*. It omits names, places, monetary amounts, ratios, contract wording, parcel identifiers and identity data, and it reproduces no scan or stamp. What remains is the problem structure, the workflow, the deliverable design and countable facts, which are sufficient to show how the work was done. A fuller walkthrough can be given orally under a confidentiality agreement, still without party information.

2 Source Material and Corpus Analysis

2.1 Composition of the Corpus

The corpus is the complete set of scans the family held. Table 2 classifies the 39 pages; Figure 1 shows the proportions. Agreements dominate, but nearly one page in five is an identity document, which is why data minimisation was a design rule from the start: identity data were read for role mapping only and never copied into a deliverable.

Table 2: Classification of the 39 scanned pages

Category	Pages	Notes
Agreements and contracts	20	Three parcel-level cooperative-construction agreements (4–5 pages each) and one larger project contract (6 pages)
Supplements and side note	3	Two supplementary instruments and one handwritten side note from 2018
Applications	2	Two demolition-and-rebuild applications, each stamped by the local collective
Identity documents	7	Front and back copies, several partly obscured by handwriting
Land registration and maps	6	Parcel maps, a land-use certificate cover, two registration forms, one relative-position map
Acceptance record	1	Completion acceptance signed by collective representatives



Share of the 39 scanned pages by document type. Other = supplements 2, applications 2, acceptance record 1, side note 1.

Figure 1: Composition of the corpus by document type

2.2 Quality Problems That Shaped the Method

Five properties of the scans drove design choices. (i) Handwritten entries and stamps overlap typed text, so machine transcription needs human confirmation. (ii) Pages are skewed or shadowed, so an unreadable field must be reported as unreadable rather than guessed. (iii) Several agreements are unfilled templates: key fields such as the number of floors and areas are blank, so the text alone cannot fix what was built. (iv) One supplementary instrument carries the landholders’ signatures but a blank investor signature line and no date, so it cannot be treated as effective. (v) Four parcels use near-identical wording, so evidence from one parcel can be wrongly attributed to another unless every claim carries a parcel reference.

2.3 What the Corpus Can and Cannot Establish

Before designing anything, each business question was tested against the corpus (Table 3). The gaps in the right-hand column became the evidence-collection list.

Table 3: Questions the corpus can and cannot answer

Question	What the documents show	Gap
Who paid for construction?	Agreements state that the investor funds it in cash	No receipts, invoices or bank records in the corpus
Who holds title?	Buildings are to be registered in the landholder’s name; no certificates present	Current registration status unknown
What happens on expropriation?	Compensation follows the area held by each side	No current expropriation or renewal notice on file
Are heirs bound?	Each agreement binds legal heirs of both sides	Identity and number of heirs not documented
Is there a split between investors?	Each agreement names one investor; two different investors appear across parcels	No document records any split or ratio
Who collected past income?	A 2018 note assigns rent for a small area to the owner during tenancy	No leases, ledgers or bank records

3 Solution Design

3.1 Overview

The solution treats the task as a document-intelligence problem with a human verification loop (Figure 2). An LLM assistant ingests the scanned records and structures them; the author checks every extracted claim against the original image before it may enter a deliverable; the outputs are graded by confidence and handed to professional review. Gaps found during verification are fed back into the structuring step instead of being papered over.

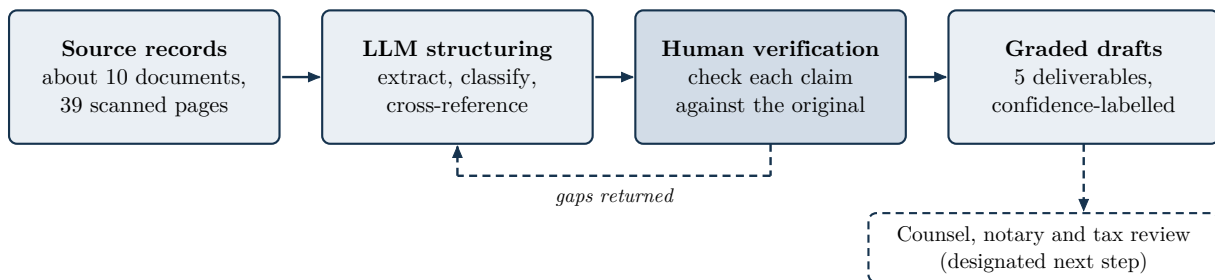


Figure 2: Workflow with a human verification loop and professional review as the final gate

3.2 Three Workstreams and Their Order

The plan splits the problem into three workstreams (Table 4). The order is deliberate: succession is verified *before* anything is signed, otherwise a later, unsigned heir could contest the whole arrangement; the management entity comes last and is explicitly a management vehicle that does not acquire title.

Table 4: Workstreams, objectives and dependencies

Workstream	Objective	Key output	Needs
1 Succession check	Identify every party entitled to sign or authorise	Party list, evidence checklist, joint-declaration outline	—
2 Contribution and entitlement	Record historical contributions and the rules for sharing future proceeds	Draft confirmation agreement with annexes	1
3 Management entity	One accountable body for collection, invoicing, upkeep and records	Entity design memo, incorporation guide	1, 2

3.3 Thirty-Day Sequencing

The roadmap converts the order into four time windows (Figure 3). Evidence collection comes first because every later document depends on it; counsel’s check of parcel status runs in parallel with party contact; the entity is incorporated only in the last window, and rent collection is moved only after a supplementary agreement is signed with each tenant.

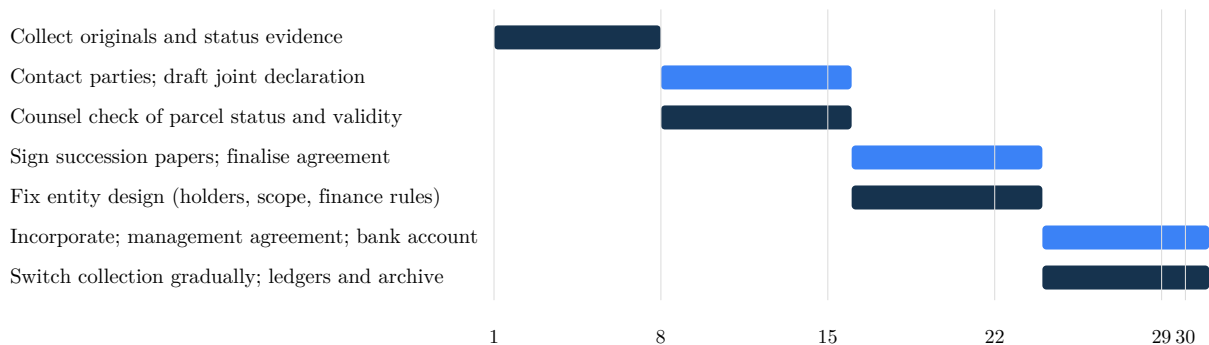


Figure 3: Thirty-day plan by window (day 1 to day 30)

3.4 Management Entity Options

Three ownership structures were compared (Table 5). The memo recommends the first when heirs are clear and cooperative, and the second while succession is unresolved, because registration of a many-holder entity stalls when any holder is undetermined.

Table 5: Ownership options for the management entity

Opt.	Structure	Advantage	Risk
A	Ownership mirrors the economic interests	Consistent with the history; simple exit and distribution paths	May be read as if the underlying asset had been contributed; blocked by any undetermined holder
B	One controlling founder; other parties hold contractual entitlements	Fast to set up; avoids many heirs as direct holders	Other side may fear loss of control; needs strong audit and consent rights
C	A holding vehicle above the entity	Suits later inheritance; flexible roles and exits	Higher initial complexity and tax and governance cost

3.5 Governance Rules and Deliberate Limits

Two lists constrain the design. **Governance:** dual approval and payment limits on the bank account; joint consent above a spending threshold; joint consent to sign, renew or end a tenancy; unanimous consent for sale, transfer of contract rights, compensation talks, pledge, guarantee and borrowing; periodic statements; an annual joint check of accounts; and a clause that no party may use the entity against another's interest. **Not yet:** do not present the entity as owner; do not promise a sale of title; do not contribute unregistered assets as capital; do not let an offshore vehicle take underlying rights; do not sign any exclusive sale, pledge, financing or irrevocable authority before due diligence.

4 AI-Assisted Implementation

4.1 Ingestion and Structuring

The LLM assistant transcribed each scan, labelled the document type, extracted the parties' roles, dates, plot descriptions and obligations, and assembled one chronology from 2008 to 2018 (Figure 4). Cross-reading the structured records produced five findings that no single file shows: (i) the four parcels each have their own agreement and cannot be treated as one arrangement; (ii) two distinct investor parties appear across them, which is consistent with a co-investment but does not document one; (iii) several template agreements leave key fields blank; (iv) one supplement is unsigned by the investor; and (v) a 2018 side note allocates rent from a small area to the owner and provides proportional sharing if the area is expropriated.

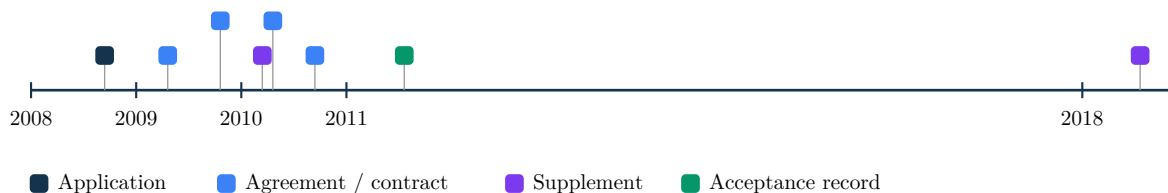


Figure 4: Eight dated instruments reconstructed from the scans, 2008–2018 (generalised types only)

4.2 Drafting the Deliverables

From the structured record the assistant drafted five documents (Table 6). Each opens with a statement that it is a working draft requiring professional review, and sections that depend on missing evidence contain blank fields rather than plausible-sounding text.

Table 6: Deliverables and their scale

Document	Content and scale
Execution roadmap	Three workstreams; 30-day plan in four time windows; 7 open information gaps
Draft confirmation agreement	13 clauses and 5 annexes covering scope, sharing rules, dispute handling and tax allocation
Succession checklist	10 questions that must be answered; 3 evidence groups; 5 risk warnings; declaration outline of 7 core terms
Entity design memo	3 ownership structures compared; governance rules; 8 pre-incorporation documents; 8 first actions; 5 things not to do yet
Incorporation guide	8 upfront decisions, 7 filing steps, 10 post-licence actions, 10 common pitfalls

4.3 Anatomy of the Key Deliverables

Confirmation agreement. Its 13 clauses are listed in Table 7. Two design choices matter more than the wording. Clause 10 restricts the document to contribution, contract rights, income sharing and management, and states that it creates, transfers or registers no title; and clause 5 requires a written agreement on price, tax burden, payment route and delivery conditions before any major transaction is signed.

Table 7: Clause topics of the draft confirmation agreement

Clauses	Topics
1–2	Project background; confirmation of the historical contribution basis (a placeholder pending evidence)
3–5	Scope of interests (income, compensation, future registration, entity interests); sharing of recurring income after costs; sale, transfer and compensation proceeds with mutual veto over unilateral disposal
6–8	Management entity as agent only; evidence and records to be annexed; succession and successors bound
9–10	Tax and cost allocation, with fault-based liability for misstatement; title and compliance boundary
11–13	Breach and remedies (return of diverted income with funding cost); dispute resolution; effectiveness and annexes

Succession checklist. Its ten questions ask, in order: the identity and status of the party; whether a will or support agreement exists; marital status at death; whether the spouse is living; the list of children, including adopted and step-children with a support relationship; whether the parents were living; whether notarisation, a court settlement or an estate division already exists; whether anyone has waived; whether any heir has died, triggering substitution; and who is authorised to sign for the group. Evidence is grouped as materials on the deceased, materials on each heir, and materials on the project entitlement.

Incorporation guide. The eight upfront decisions are positioning, name, ownership, capital, legal representative, finance and tax roles, registered address and business scope. Capital is kept moderate because contribution falls due within five years; the legal representative must be a person able to sign for years, not a nominal one; and the address must be real and reachable, with a warning against unsolicited offers of “official” registered addresses.

4.4 Verification and Confidence Grading

Verification was manual and source-based: every statement in a draft was traced to a page of the scans, or it was downgraded. Table 8 shows the grading scheme with generalised examples.

Table 8: Confidence grades used across all deliverables

Grade	Meaning	Generalised example
Confirmed	Directly supported by an original document	Buildings are registered, if at all, in the landholder’s name
To be verified	Plausible; required evidence not yet in hand	Succession status; who received past income; two investors and their relationship
Hypothesis	Appears only in a draft, with no source	A proposed fixed split of interests

5 Judgement and Risk Reasoning

5.1 A Judgement Call on Evidence

An AI-drafted agreement described the proposed split of interests as *already confirmed*. When each such statement was traced to the scans, no document supported it: each original agreement names a single investor and none records a split between investors. The claim was downgraded to *hypothesis*, the draft wording was changed from “confirms” to “proposes”, and an evidence list (payment records, receipts, witness statements) was added as the route to upgrading it. The lesson is that fluent output is not evidence; the value added was the verification loop, not the drafting.

5.2 Designing Against Signature Gaps

The succession checklist encodes a rule that prevents a common failure: a document signed by fewer people than the law requires. One heir’s signature is not enough unless every other heir has authorised that person in writing; a spouse is normally a key signatory; parents may be heirs if living; and an heir who has died opens a substitution or transfer question that cannot be ignored. The simplest signing set, valid only if no will, no missing heir and no living parent are found, is the spouse and the children, or a written authorisation from all of them that states its scope and confirms that the party’s entitlement is a share, not the whole.

5.3 Risk Register

Table 9 collects the risks that the deliverables are designed against, drawn mainly from the incorporation guide’s list of pitfalls and the checklist’s warnings.

Table 9: Risk register with designed mitigations

Risk	Consequence	Designed mitigation
An unsigned or missing heir	Later challenge to the whole arrangement	Verify heirs first; require full list or written authorisation
Split of interests treated as fact	Dispute when evidence is asked for	Grade as hypothesis; list evidence to collect
Entity presented as owner	Enlarges title risk; misleads counterparties	Manager-only positioning in every document
Contract, invoice and collection entity differ	Tax explanation becomes difficult	Sign supplementary tenancy agreements before switching collection
Unregistered assets contributed as capital	Invalid or challengeable contribution	Explicit prohibition in the entity memo
Company and personal accounts mixed	Loss of audit trail	Dedicated collection account; dual approval
Income undeclared or under-declared	Tax exposure and penalties	Tax registration and ledgers from day one
Template governance text	No protection for minority holders	Written unanimous-consent list and open books
LLM overstatement	Draft accepted as fact	Source tracing; confidence grades; blank fields instead of guesses
Exposure of identity data	Privacy harm	Role mapping only; identity data kept out of all deliverables

6 Outcomes, Limitations and Reflection

6.1 Results at a Glance

Table 10: Countable results

Item	Count	Item	Count
Documents and records analysed	~10	Draft agreement clauses	13
Scanned pages	39	Draft agreement annexes	5
Parcels distinguished	4	Must-answer succession questions	10
Period covered	2008–2018	Open information gaps listed	7
Draft deliverables	5	Ownership options compared	3
Workstreams / plan windows	3 / 4	Risks with mitigations	10

The scattered paper record became one structured chronology and five draft documents that turn an open-ended family issue into a sequenced, reviewable plan. It states who must be identified, what evidence is missing, in which order actions should happen and where professional advice is mandatory. It also produced a concrete list of information gaps that the family can now collect, in place of a vague sense that “the papers are incomplete”.

6.2 Limitations

This is analysis and design work at draft stage. No agreement has been signed and no entity registered, and no professional review is claimed. Income figures that appear in working drafts were not tied to leases or bank records and are excluded here. The LLM assistant is described generically because the source files do not record which product produced each draft. Effort saved is not quantified, since no baseline was measured. Because the source scans are not reproduced, a reader cannot re-check the findings from this document alone.

6.3 Transferable Lessons for AI Solution Work

Grade every claim. A three-level scheme makes an AI draft usable without letting it pose as evidence. **Order the work by dependency.** The sequence succession, then agreement, then entity is what turns a pile of drafts into a plan. **Bound the scope in writing.** Each document says what it does not do. **Feed gaps back.** Missing evidence returns to the structuring step and becomes a collection task. **Minimise personal data.** Roles were mapped, identities were not carried.

6.4 Who Did What

Table 11 separates the roles so that a reader can see exactly where the author’s contribution starts and stops.

Table 11: Division of work

Role	Did	Did not do
Author	Collected originals; directed the assistant; verified every claim against the scans; set grades and sequence; coordinated with family members	Give legal, tax or notarial advice; sign or register anything
LLM assistant	Transcribed, classified and cross-referenced records; drafted five working documents	Decide what counts as evidence; supply facts not in the scans
Licensed professionals (designated)	Review the agreement and entity design; handle notarisation, tax and registration	Not yet engaged at the time of writing

Role	Did	Did not do
Family members and collective	Supply originals and status information; sign when the time comes	Not asked to sign anything at draft stage

6.5 Evidence Needed to Upgrade Each Open Item

Every item graded below *confirmed* has a defined route to upgrade. Table 12 lists what would move each one, so the next step for the family is a collection task, not a debate.

Table 12: Upgrade path for open items

Open item	Evidence that would upgrade it	Held by
Who funded construction	Payment vouchers, contractor settlements, receipts, bank statements for each parcel	Family archive; contractors; banks
Split between two investors	Contemporaneous messages, remittance records, witness statements, any written note	Both families; witnesses
Current registration status	Parcel-level status from the local registry or the collective; confirmation of any certificate	Collective; land authority
Succession of the deceased party	Death and household-cancellation records, kinship proofs, any will, notarised or court settlement	Family; notary; court
Rent history and current tenants	Leases, deposit receipts, rent ledgers, invoices, tax filings	Tenants; accountant
Compensation or renewal prospects	Written notices from the authority; collective resolutions	Collective; authority
Validity of the unsigned supplement	Signed and dated original, or written waiver by the investor	Investor and landholder

6.6 Decisions That Depend on the Open Items

Three decisions are deliberately not taken in the current drafts because each depends on an open item. **Which ownership option for the entity** depends on the succession result, since a single undetermined holder blocks a many-holder structure. **Whether to switch rent collection** depends on the tenancy evidence and on each tenant signing a supplementary agreement. **Whether to negotiate any sale or compensation** depends on registration status and on the agreed routing of proceeds, and requires unanimous consent in the governance rules. Writing these dependencies down keeps the analysis from drifting into advice it cannot support.

6.7 Next Steps

(i) Collect the succession evidence listed in the checklist; (ii) have counsel review the confirmation agreement and the entity design; (iii) replace each hypothesis with documentary support or remove it; (iv) only then move income collection to the management entity, using a supplementary agreement with existing tenants.

6.8 Verification Offered

On request and under a confidentiality agreement, the following can be shown on screen without party information: the structure and field design of the chronology; the five draft documents with all names and figures masked; the grading of a sample of findings against page references; and the succession checklist as a blank template. Original scans and identity papers are not shown. This gives a reviewer a way to test the method and the honesty of the grading without exposing the underlying family matter.